

MATHS ASSIGNMENT (NO QUESTION TO BE LEFT FOR OPTIONS ATTEMPT ALL THE QUESTIONS)

SUBMISSION DATE : ON OR BEFORE NOVEMBER 15



Reg. No.

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BCACACN 201

Second Semester B.C.A. Degree Examination, September 2022
(NEP – 2020) (2021-22 Batch Onwards)
DISCRETE MATHEMATICAL STRUCTURES (DSCC)

Time : 2 Hours

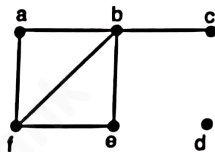
Max. Marks : 60

Note : Answer any six questions from Part – A and one full question from each Unit in Part – B.

PART – A

(6×2=12)

- Write which of these is a tautology or a fallacy i) $p \vee \neg p$ ii) $p \wedge \neg p$.
- Write the power set of $A = \{1, 2\}$.
- Define partial order relation. Give an example.
- What is Pigeonhole principle ? Give an example.
- What is the probability that when two dice are rolled, the sum of the numbers on the two dice is 7 ?
- Define prime and composite number.
- Define degree of a vertex, with an example.
- Find isolated and pendant node in the graph :



PART – B

UNIT – I

- Using truth table, show that $(p \wedge q) \Rightarrow p$ and $p \Rightarrow (p \vee q)$ are both tautologies, where p and q are any two statements.
- $A = \{3, 4, 5, 17\}$, $B = \{1, 2, 3\}$, $C = \{x | x \text{ is an integer and } 0 \leq x \leq 5\}$ write $A \cup B$, $A \cup C$.
- $R = \{<1, 2>, <3, 4>, <2, 2>\}$, $S = \{<4, 2>, <2, 5>, <3, 1>, <1, 3>\}$. Write $R \circ S$ and $S \circ S$.
(4+4+4)
P.T.O.

CACACN 201

3. a) With the help of truth table, prove that $p \Rightarrow (q \wedge r) \equiv (p \Rightarrow q) \wedge (p \Rightarrow r)$.
 b) $R = \{ \langle 1, 1 \rangle, \langle 1, 2 \rangle, \langle 1, 4 \rangle, \langle 2, 1 \rangle, \langle 2, 2 \rangle, \langle 2, 3 \rangle, \langle 3, 2 \rangle, \langle 3, 3 \rangle, \langle 4, 2 \rangle, \langle 4, 4 \rangle \}$
 Construct relation matrix of R and draw digraph of R.
 c) $A = \{1, 2, 3\}$, $B = \{1, 2, 5, 7, 9\}$. Write $A - B$, $A + B$, $A \cup B$, $A \cap B$. (4+4)

UNIT – II

4. a) Draw the Hasse diagram of the set A, under the partial ordering relation "divides" and indicate those which are totally ordered. $A = \{1, 2, 3, 6, 12\}$.
 b) A multiple-choice test contains 10 questions. There are four possible answers for each question. A) In how many ways can a student answer the questions on the test if the student answers every question? B) In how many ways can a student answer the questions on the test if the student can leave answers blank?
 c) Show that functions $f(x) = x^3$ and $g(x) = x^{1/3}$ for $x \in \mathbb{R}$ are inverse of one another. (4+4)
5. a) Let $X = \{1, 2, 3\}$ f, g, h and s are the functions from X to X given by
 $f = \{(1, 2), (2, 3), (3, 1)\}$, $h = \{(1, 1), (2, 2), (3, 1)\}$, $g = \{(1, 2), (2, 1), (3, 3)\}$,
 $s = \{(1, 1), (2, 2), (3, 3)\}$. Find $f \circ g$, $g \circ f$, $s \circ s$ and $f \circ s$.
 b) A computer company receives 350 applications from computer graduates for a job planning a line of new web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business?
 c) Let $S = \{1, 2, 3, 4, 5\}$. List all the 3-permutations and 3-combinations of S. (4+4)

UNIT – III

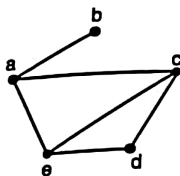
6. a) Find the probability that a hand of five cards in poker contains four cards of one kind.
 b) What is the conditional probability that a family with two children has two boys, given they have at least one boy?
 c) Find the greatest common divisor of 414 and 662 using the Euclidean algorithm. (4+4)



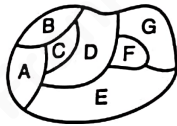
7. a) Are the random variables x_1 and x_2 independent, if the sum of the numbers that appear when a pair of fair dice is rolled ?
- b) Use mathematical induction to prove that $2n < n!$ for every integer n with $n \geq 4$. (6+6)

UNIT – IV

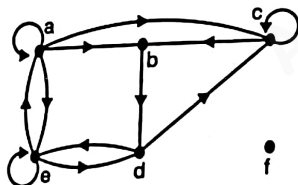
8. a) Write adjacency list and adjacency matrix to describe the simple graph given in the figure.



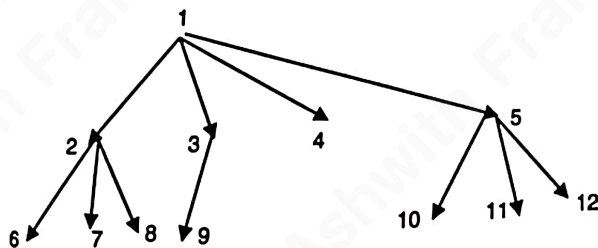
- b) Construct the dual graph for the given map. (6+6)



9. a) Find the in-degree and out degree of each vertex in the graph G with directed edges as shown in the figure.



- b) Convert the following tree into a binary tree. (6+6)



Part - A :

1. a. Write which of these is a tautology or a fallacy

(i) $P \vee \neg P$

=

P	$\neg P$	$P \vee \neg P$
T	F	T
F	T	T

(ii) $P \wedge \neg P$

=

P	$\neg P$	$P \wedge \neg P$
T	F	F
F	T	F

$\therefore P \vee \neg P$ is a tautology
 $P \wedge \neg P$ is a fallacy

b. Write the power set of $A = \{1, 2\}$

= $A = \{1, 2\}$

$P(A) = 2^n$

$= 2^2$

= 4

c. Define partial order relation. Give an example.

= It is a binary relation, R in a set P is called a partial order relation or a partial ordering in P iff R is reflexive, antisymmetric, and transitive. It is denoted by $\leq$ (less than or equals to)

$(P, \leq)$ if the ~~is~~ is a partial ordering on P , then the ordered pair $(P, \leq)$ is called partially ordered set or a POSET (Partially Ordered Set)

d. What is Pigeonhole principle? Give an example.

= Pigeonhole principle states that if k is positive integer and $k+1$ or more objects are placed into k boxes, then there is at least one box containing two or more of the objects.

Ex: Among any group of 367 people, there must be at least two with the same birth day, because there are only 366 possible birth days.

e. What is the probability that when two dice are rolled, the sum of the numbers on the two dice is 7?

= Finding the probability of getting a sum of 7 when the two dice are rolled. Total number of outcomes $6 \times 6 = 36$. Favorable outcomes (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), and (6, 1).
 $6/36 = 1/6$

f. Define prime and composite number.

= A prime number is a number greater than 1 that has exactly two distinct factors, 1 and itself.

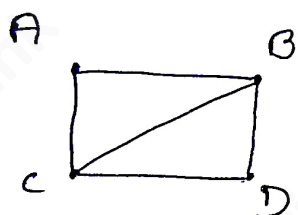
Ex: 2, 3, 5, 7, 11, 13, 17, ...

A composite number is a number greater than 1 that has more than two factors, meaning it can be divided exactly by numbers other than 1 and itself.

Ex: 4, 6, 8, 9, 10, 12, ...

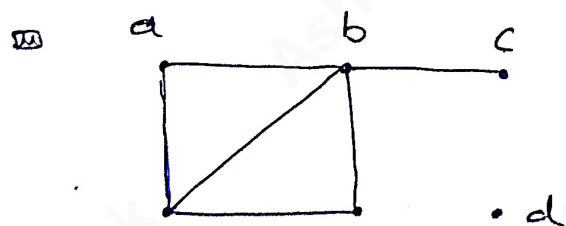
g. Define degree of a vertex, with an example.

= The degree of a vertex in a graph is the number of edges incident to (connected to) that vertex.



<u>Vertex</u>		<u>Degree</u>
A	-	2
B	-	3
C	-	3
D	-	2

h. Find isolated and pendent node in the graph.



= d is Isolated node (degree of 0)
c is Pendent node (leaf node)

Part - B :-

2. a. Using truth table, show that $(p \wedge q) \Rightarrow p$ and $p \Rightarrow (p \vee q)$ are both tautologies, where p and q are any two statements.

① $(p \wedge q) \Rightarrow p$

p	q	$p \wedge q$	$(p \wedge q) \Rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

② $p \Rightarrow (p \vee q)$

p	q	$p \vee q$	$p \Rightarrow (p \vee q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

$\therefore$ Hence, both the statements are true
 $\therefore$ Tautologies are proved.

b. $A = \{3, 4, 5, 17\}$, $B = \{1, 2, 3\}$, $C = \{x \mid x \text{ is an integer and } 0 \leq x \leq 5\}$ write $A \cup B, A \cup C$.

$$A = \{3, 4, 5, 17\}$$

$$B = \{1, 2, 3\}$$

$$C = \{0, 1, 2, 3, 4, 5\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 17\}$$

$$A \cup C = \{0, 1, 2, 3, 4, 5, 17\}$$

C. $R = \{(1, 2), (3, 4), (2, 2)\}$, $S = \{(4, 2), (2, 5), (3, 1), (1, 3)\}$. Write $R \circ S$ and $S \circ R$.

$$R = \{(1, 2), (3, 4), (2, 2)\}$$

$$S = \{(4, 2), (2, 5), (3, 1), (1, 3)\}$$

$$(4, 2) \in S$$

$$(2, 2) \in R$$

$$= (4, 2)$$

$$(3, 4) \in S$$

$$(1, 2) \in R$$

$$= (3, 2)$$

$$(1, 3) \in S$$

$$(3, 4) \in R$$

$$= (1, 4)$$

$$R \circ S = \{(4, 2), (3, 2), (1, 4)\}$$

$$(4, 2) \in S$$

$$(2, 5) \in S$$

$$= (4, 5)$$

$$(3, 1) \in S$$

$$(1, 3) \in S$$

$$= (3, 3)$$

$$(1, 3) \in S$$

$$(3, 1) \in S$$

$$= (1, 1)$$

$$S \circ S = \{(4, 5), (3, 3), (1, 1)\}$$

$$\therefore R \circ S = \{(4, 2), (3, 2), (1, 4)\}$$

$$S \circ S = \{(4, 5), (3, 3), (1, 1)\}$$

3. a. With the help of truth table, prove that

$$p \Rightarrow (q \wedge r) \equiv (p \Rightarrow q) \wedge (p \Rightarrow r)$$

=

p	q	r	$q \wedge r$	$p \Rightarrow (q \wedge r)$
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
T	F	F	F	F
F	T	T	T	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

p	q	r	$p \Rightarrow q$	$p \Rightarrow r$	$(p \Rightarrow q) \wedge (p \Rightarrow r)$
T	T	T	T	T	T
T	T	F	F	F	F
T	F	T	F	F	F
T	F	F	F	F	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	F	F
F	F	F	T	T	T

$\therefore$ Hence, $p \Rightarrow (q \wedge r) \equiv (p \Rightarrow q) \wedge (p \Rightarrow r)$ proved.

b. $R = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 2), (4, 4)\}$. Construct relation matrix of R and draw diagram of R .

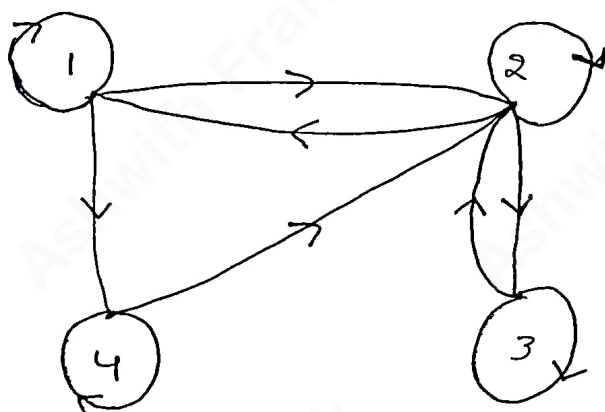
=

(diagram)

$$= M_R$$

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Graph :



R

C. $A = \{1, 2, 3\}$, $B = \{1, 2, 5, 7, 9\}$. Write $A+B$, $A+B$, $A \cup B$, $A \cap B$.

$$= A = \{1, 2, 3\}$$

$$B = \{1, 2, 5, 7, 9\}$$

$$A-B = \{3\}$$

$$A+B = \{1, 2, 3, 5, 7, 9\}$$

$$A \cup B = \{1, 2, 3, 5, 7, 9\}$$

$$A \cap B = \{1, 2\}$$

4. a. Draw the Hasse diagram of the set A , under the partial ordering relation "divides" and indicate those which are totally ordered. $A = \{1, 2, 3, 6, 12\}$

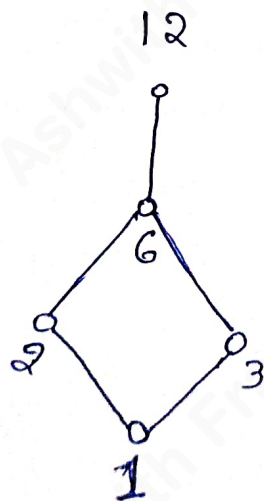
$\{$
 ~~$(1, 1)$~~ , $(1, 2)$, $(1, 3)$, $(1, 6)$, $(1, 12)$,
 ~~$(2, 1)$~~ , ~~$(2, 2)$~~ , $(2, 3)$, $(2, 6)$, $(2, 12)$,
 ~~$(3, 1)$~~ , $(3, 2)$, ~~$(3, 3)$~~ , $(3, 6)$, $(3, 12)$,
 $(6, 1)$, $(6, 2)$, $(6, 3)$, ~~$(6, 6)$~~ , $(6, 12)$,
 $(12, 1)$, $(12, 2)$, $(12, 3)$, $(12, 6)$, ~~$(12, 12)$~~ $\}$

All ordered pairs with $a|b$

$\{$
 $(1, 1)$, $(2, 2)$, $(3, 3)$, $(6, 6)$, $(12, 12)$,
 $(1, 2)$, $(1, 3)$, $(1, 6)$, $(1, 12)$,
 $(2, 6)$, $(2, 12)$, $(3, 6)$, $(3, 12)$, $(6, 12)$ $\}$

Final Hasse pairs:

$\{$
 $(1, 2)$, $(1, 3)$, $(2, 6)$, $(3, 6)$, $(6, 12)$ $\}$



b. A multiple-choice test contains 10 questions. There are four possible answers for each question.

a. In how many ways can a student answer the questions?

= There are 10 questions and 4 possible answers for each question.

$$\text{Total ways} = 4^{10}$$

$$4^{10} = (4^5)^2 = (1024)^2 = \underline{\underline{1,048,576}}$$

b. In how many ways can a student answer the questions on the test if the student can leave answer blank?

= If the student can leave a question blank, that means each question now has 5 possible choices (4 answers + 1 blank).

$$\text{Total ways} = 5^{10}$$

$$5^{10} = (5^5)^2 = (3125)^2 = \underline{\underline{9,765,625}}$$

c. Show that functions $f(x) = x^3$ and $g(x) = x^{1/3}$ for $x \in \mathbb{R}$ are inverse of one another.

= Using inverse function theorem 2 and 2, we prove $f(x)$ and $g(x)$ inverse of one another. Therefore,

$$f \circ g(x) = f(g(x)) = x \quad \text{--- (1)}$$

$$g \circ f(x) = g(f(x)) = x \quad \text{--- (2)}$$

$$\textcircled{1} \quad f(g(x))$$

$$f(x^{1/3})$$

$$x^{1/3} \cdot x^3$$

$$\underline{\underline{x' = x}}$$

$$f \circ g(x) = f(g(x)) = \underline{\underline{x}}$$

$$\textcircled{2} \quad g(f(x))$$

$$g(x^3)$$

$$x^3 \cdot x^{1/3}$$

$$\underline{\underline{x' = x}}$$

$$g \circ f(x) = g(f(x)) = \underline{\underline{x}}$$

5. a. Let $X = \{1, 2, 3\}$ & f, g, h and s are the functions from X to X given by
 $f = \{(1, 2), (2, 3), (3, 1)\}$, $h = \{(1, 1), (2, 2), (3, 3)\}$,
 $g = \{(1, 2), (2, 1), (3, 3)\}$, $s = \{(1, 1), (2, 2), (3, 3)\}$. Find $f \circ g$, $g \circ f$, $s \circ s$ and $f \circ s$.

= Function of g :-

$$g(1) = 2$$

$$g(2) = 1$$

$$g(3) = 3$$

Function of h :

$$h(1) = 1$$

$$h(2) = 2$$

$$h(3) = 1$$

Function of β :

$$\beta(1) = 2$$

$$\beta(2) = 3$$

$$\beta(3) = 1$$

Function of s :

$$s(1) = 1$$

$$s(2) = 2$$

$$s(3) = 3$$

$\beta \circ g$:

$$\beta(g(1)) = \beta(2) = 3 \Rightarrow (1, 3)$$

$$\beta(g(2)) = \beta(1) = 2 \Rightarrow (2, 2)$$

$$\beta(g(3)) = \beta(3) = 1 \Rightarrow (3, 1)$$

$$\beta \circ g = \{(1, 3), (2, 2), (3, 1)\}$$

$g \circ \beta$:

$$g(\beta(1)) = g(2) = 1 \Rightarrow (1, 1)$$

$$g(\beta(2)) = g(3) = 3 \Rightarrow (2, 3)$$

$$g(\beta(3)) = g(1) = 2 \Rightarrow (3, 2)$$

$$g \circ \beta = \{(1, 1), (2, 3), (3, 2)\}$$

$g \circ s$:

$$g(s(1)) = g(1) = 2 \Rightarrow (1, 2)$$

$$g(s(2)) = g(2) = 1 \Rightarrow (2, 1)$$

$$g(s(3)) = g(3) = 3 \Rightarrow (3, 3)$$

$$g \circ s = \{(1, 2), (2, 1), (3, 3)\}$$

Sos :-

$$s(s(1)) = s(1) = 1 \Rightarrow (1, 1)$$

$$s(s(2)) = s(2) = 2 \Rightarrow (2, 2)$$

$$s(s(3)) = s(3) = 3 \Rightarrow (3, 3)$$

$$s \circ s = \{(1, 1), (2, 2), (3, 3)\}$$

g \circ s :-

$$g(s(1)) = g(1) = 2 \Rightarrow (1, 2)$$

$$g(s(2)) = g(2) = 3 \Rightarrow (2, 3)$$

$$g(s(3)) = g(3) = 1 \Rightarrow (3, 1)$$

$$g \circ s = \{(1, 2), (2, 3), (3, 1)\}$$

b. A computer company receives 350 applications from computer graduates for a job planning a line of new servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business?

$$= \text{Total applicants} = 350$$

$$\text{Majored in Computer Science (CS)} = 220$$

$$\text{Majored in Business (B)} = 147$$

$$\text{Majored in both CS and Business} = 51$$

$$|CS \cup B| = |CS| + |B| - |CS \cap B|$$

$$|CS \cup B| = 220 + 147 - 51 = 316$$

$$\text{Neither} = \text{Total} - |\text{CS} \cup \text{B}|$$

$$\text{Neither} = 350 - 316 = 34$$

$\therefore$ 34 applicants majored neither in computer science nor in business.

c. Let $S = \{1, 2, 3, 4, 5\}$. List all the 3 permutations and combinations of S .

= The 3-permutations of S :-

The number of k -permutations from a set of n elements is given by the

$$\text{formula } P(n, k) = \frac{n!}{(n-k)!}$$

$$n = 5$$

$$k = 3$$

$$P(5, 3) = \frac{5!}{(5-3)!}$$

$$= \frac{5!}{2!}$$

$$= \frac{120}{2}$$

$$= \underline{\underline{60}}$$

The number of k -combinations from a set of n elements is given by the

$$\text{formula } C(n, k) = \frac{n!}{k!(n-k)!}$$

$$n = 5$$

$$k = 3$$

$$C(5, 3) = \frac{5!}{3!(5-3)!}$$

$$= \frac{5!}{3!2!} = \frac{120}{6 \cdot 2} = \underline{\underline{10}}$$

3-combinations: $\therefore (10)$

$\{(1,2,3), (1,2,4), (1,2,5), (1,3,4), (1,3,5),$
 $(1,4,5), (2,3,4), (2,3,5), (2,4,5), (3,4,5)\}$

3-permutations: $\therefore (60)$

$\{(1,2,3), (1,3,2), (2,1,3), (2,3,1), (3,1,2), (3,2,1),$
 $(1,2,4), (1,4,2), (2,1,4), (2,4,1), (4,1,2), (4,2,1),$
 $(1,2,5), (1,5,2), (2,1,5), (2,5,1), (5,1,2),$
 $(5,2,1), (1,3,4), (1,4,3), (3,1,4), (3,4,1),$
 $(4,1,3), (4,3,1), (1,3,5), (1,5,3), (3,1,5),$
 $(3,5,1), (5,1,3), (5,3,1), (1,4,5), (1,5,4),$
 $(4,1,5), (4,5,1), (5,1,4), (5,4,1), (2,3,4),$
 $(2,4,3), (3,2,4), (3,4,2), (4,2,3), (4,3,2),$
 $(2,3,5), (2,5,3), (3,2,5), (3,5,2), (5,2,3),$
 $(5,3,2), (2,4,5), (2,5,4), (4,2,5), (4,5,2),$
 $(5,2,4), (5,4,2), (3,4,5), (3,5,4), (4,3,5),$
 $(4,5,3), (5,3,4), (5,4,3)\}$

6. a. Find the probability that a hand of five cards in poker contains four cards of one kind.

= Given 5 cards of poker hand drawn from a deck of 52 cards.

Eg: 4 kings + 1 other card.

$$\therefore \text{Total hands} = {}^{52}C_5 = \frac{52!}{5!}$$

$$= \frac{52!}{5!(52-5)!}$$
$$= 25,989,600$$

1. Choose the rank for the four of a kind.
There are 13 possible ranks (A, 2, 3, ..., K)
2. Choose all 4 cards of that rank.
There are 4 cards of that rank, and we must take all 4.
3. Choose the remaining (fifth) card.
It must be of a different rank (not the same as the four-of-a-kind) so we can choose 1 card from the remaining 48 cards.

b. What is the conditional probability that a family with two children has two boys, given they have at least one boy?

= Step 1 ∴ list all possible gender combinations for two children.

Each child can be a Boy (B) or Girl (G) so, the possible outcomes (assuming equal probability and independence) are, BB, BG, GB, GG.

Each outcome has probability.

Step 2 ∴ Given condition at least one boy so the possible remaining outcomes BB, BG, GB, GG.

Probability is $\frac{1}{3}$.

Step 3 ∴ Find the probability that the family has 2 boys (BB). Only one of those (BB) satisfies two boys. So, $P(2 \text{ boys} | \text{at least 1 boy}) = \frac{P(BB)}{P(BB) + P(BG) + P(GB)} = \frac{1}{3}$

c. Find the GCD of 414 and 662 using the Euclidean algorithm.

=

$$\begin{array}{r}
 \begin{array}{r}
 1 \\
 \hline
 414 \overline{) 662} \\
 \underline{414} \\
 248
 \end{array} \\
 \begin{array}{r}
 1 \\
 \hline
 248 \overline{) 414} \\
 \underline{248} \\
 166
 \end{array} \\
 \begin{array}{r}
 1 \\
 \hline
 166 \overline{) 248} \\
 \underline{166} \\
 82
 \end{array} \\
 \begin{array}{r}
 2 \\
 \hline
 82 \overline{) 166} \\
 \underline{164} \\
 2
 \end{array} \\
 \begin{array}{r}
 41 \\
 \hline
 2 \overline{) 82} \\
 \underline{82} \\
 0
 \end{array}
 \end{array}$$

$$a = bq + r$$

$$662 = 414 \cdot 1 + 248$$

$$414 = 248 \cdot 1 + 166$$

$$248 = 166 \cdot 1 + 82$$

$$166 = 82 \cdot 2 + 2$$

$$82 = 2 \cdot 41 + 0$$

$$\therefore \text{GCD}(414, 662) = 2$$

because 2 is the last non-zero remainder.

7. a. Are the random variables x_1 and x_2 independent, if the sum of the numbers that appear when a pair of dice is rolled?

= Let x_1 be the number that appears on the first die, and x_2 be the number that appears on the second die. The sample space for a single die is $\{1, 2, 3, 4, 5, 6\}$. Since the dice are fair, the probability of rolling any number on a single die is $\frac{1}{6}$. The random variables x_1 and x_2 are independent because the outcome of one die does not affect

the outcome of the other. The sum of the numbers is a new random variable, let's call it $S = x_1 + x_2$.

The question asks if the random variables x_1 and x_2 are independent if the sum of the numbers is considered. The independence of x_1 and x_2 is a fundamental property of rolling two fair dice. The sum S is a function of these two independent variables, but this does not change their individual independence. The outcome of the first die (x_1) is still not influenced by the outcome of the second die (x_2), and vice versa.

∴ Yes, the random variables x_1 and x_2 are independent. The outcome of one die roll does not affect the outcome of the other, regardless of the sum of the numbers.

- b. Use mathematical induction to prove that $2^n < n!$ for every integer n with $n \geq 4$.

= Step 1:

$$2^n < n! = \text{Let } n \text{ be } 4$$

$$2^4 < 4! = 2 \times 2 \times 2 \times 2 < 4 \times 3 \times 2 \times 1$$

$$= \underline{16 < 24}$$

Step 2:

$$\text{Assume } n = k \\ 2^k < k!$$

Step 3:

$$2^{k+1} < (k+1)!$$

$$\text{LHS: } 2^{k+1} = 2^{m+n} = 2^m \cdot 2^n$$

$$= 2^{k+1} = 2^k \cdot 2^1$$

$$(k+1)! = (k+1) \cdot k!$$

$$\text{because } n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 2 \times 1$$

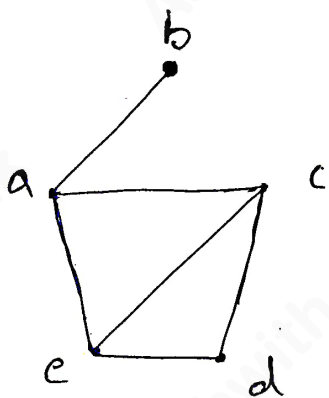
$$\therefore (k+1)! = (k+1) \cdot k \cdot (k-1) \cdot (k-2) \cdot \dots \cdot 2 \times 1$$

$$= k \cdot (k-1) \cdot (k-2) \cdot \dots \cdot 2 \times 1 \rightarrow k!$$

$$\therefore (k+1)! = (k+1) \cdot k!$$

By inductive hypothesis $2^k < k!$

8. a. Write adjacency list and adjacency matrix to describe the simple graph given in the figure.

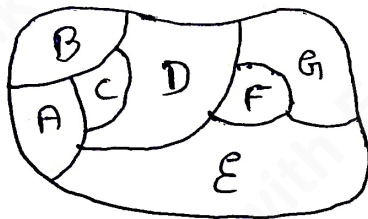


<u>Vertex</u>	<u>Adjacency Vertex</u>
a	b, c, e
b	a
c	a, e, d
d	c, e
e	a, c, d

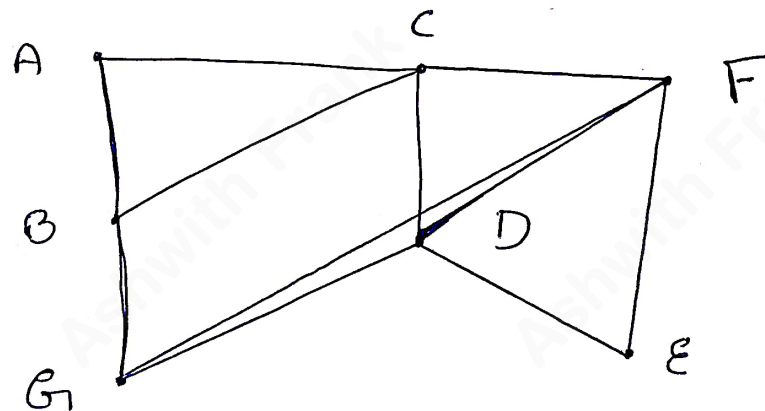
Adjacency matrix $\therefore$

	a	b	c	d	e
a	0	1	1	0	1
b	1	0	0	0	0
c	1	0	0	1	1
d	0	0	1	0	1
e	1	0	1	1	0

b. Construct the dual graph for the given map.

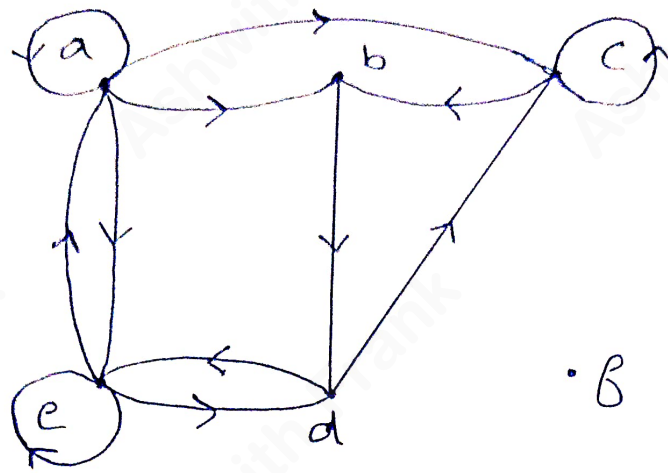


$$= A - G = \underline{\underline{7}}$$



9. a. Find the in-degree and out-degree of each vertex in the graph G_1 with directed edges as shown in the figure.

(diagram)



G₁

• β

= Indegree denoted by (-) minus.
Outdegree denoted by (+) plus.

Degree of G₁:-

$$\text{deg} - (a) = 2$$

$$\text{deg} + (a) = 3$$

$$\text{deg} - (b) = 1$$

$$\text{deg} + (b) = 1$$

$$\text{deg} - (c) = 3$$

$$\text{deg} + (c) = 2$$

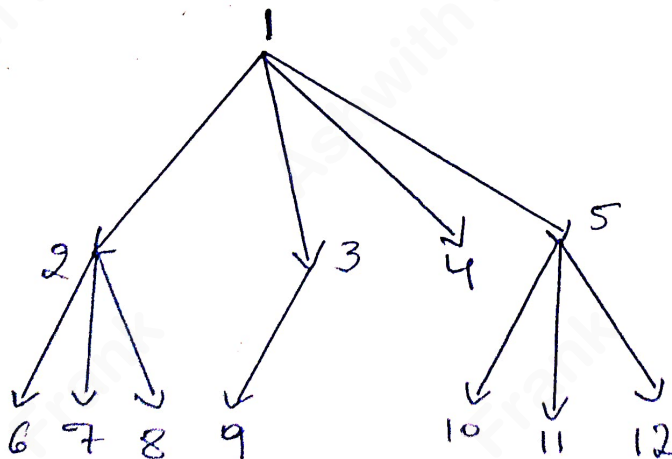
$$\text{deg} - (d) = 2$$

$$\text{deg} + (d) = 2$$

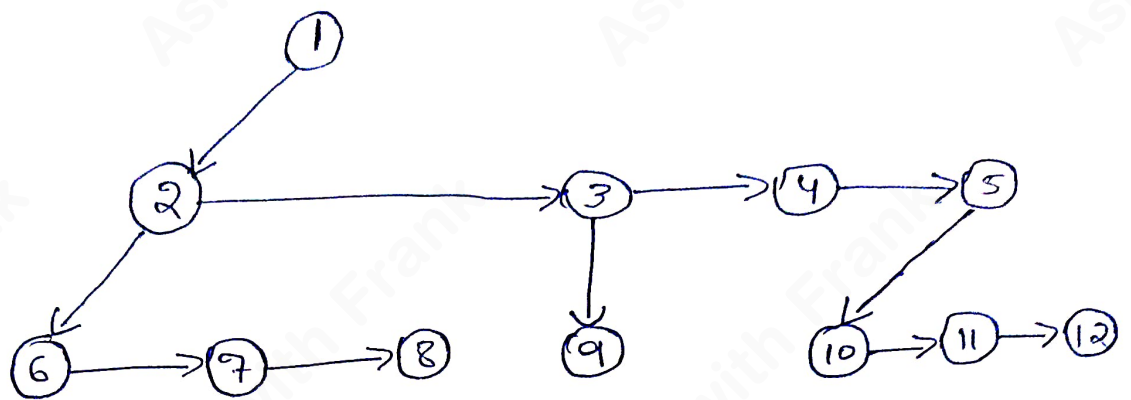
$$\text{deg} - (e) = 3$$

$$\text{deg} + (e) = 3$$

b. Convert the following tree into a binary tree.



= Step 1:-



Step 2 :-

